

Analytical Vector Stores for Scientific Research

Motivation, Research Questions, and an Early-Access Programme

Gamakon UltraDim — Research Programme

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Abstract

Scientific datasets are increasingly represented as high-dimensional vectors: molecular fingerprints, genomic and epigenomic profiles, spectra, images, simulation states, and observational matrices. Vector databases provide useful similarity retrieval for moderate-dimensional embeddings, but retrieval alone is not an analytical workflow. A scientist also needs to define cohorts, compare populations, detect structure at several scales, assess the stability of that structure, relate it to external variables, and preserve enough provenance to repeat the analysis.

We use the term *analytical vector store* for a system in which persistent vector storage, similarity retrieval, population-level analysis, and validation operate on a common corpus. Our research asks whether compact representations that make ultra-high-dimensional retrieval feasible can also support clustering, hierarchical segmentation, recommendation and matrix completion, density analysis, and visual exploration without requiring a separate computational substrate for each task. The programme is particularly concerned with wide sparse data, for which nominal dimension may reach millions while each observation occupies only a small support.

The objective is not to replace domain statistics or establish biological meaning from geometry alone. Approximate computation must be evaluated against appropriate references; exploratory structure must be tested for stability and against external labels or outcomes; and visual separation must not be treated as validation. This document sets out the scientific motivation, present scope, evaluation principles, and open questions of the programme. It also describes an academic early-access scheme through which research groups can test these ideas on scientifically interpretable workloads. Method classes are described at the level required to assess the programme; implementation recipes and unpublished construction details are outside the scope of this overview.

1 The research problem

A vector is useful because it makes a comparison operational. Once an observation has coordinates and a metric, one can ask for its neighbours, place it in a cohort, or compare it with a reference population. This abstraction is now common in machine learning, but it is older and broader than embedding models: sparse incidence vectors, molecular fingerprints, assay profiles, and discretised physical states are all vector representations.

Most vector stores concentrate on one operation: approximate nearest-neighbour retrieval. Graph indexes such as HNSW provide effective retrieval when the stored dimension and distance computation are moderate [1]. Scientific analysis, however, rarely ends with a neighbour list. Typical questions include:

- Does a sample occupy a dense, isolated, or transitional region?
- Which population structure is stable across representations and seeds?
- At what scale does a broad cohort separate into reproducible subgroups?
- Which measured variables distinguish those subgroups?
- Does geometric structure correspond to an independent biological, chemical, or clinical variable?
- Can the complete chain from source observation to analytical result be repeated after the corpus changes?

Answering these questions usually requires several disconnected systems: one for vector search, another for clustering, separate scripts for null models and enrichment, and a further export for visualisation. Each transition may change the representation, omit observations, or introduce parameters that are not recorded with the result. At high dimension, it may also duplicate the most expensive part of the computation.

Our central research question is:

Can a vector store become a reproducible analytical environment for ultra-high-dimensional scientific data, while retaining explicit controls on approximation and scientific interpretation?

This is a systems question and a statistical question. A system must make the operations feasible. An analysis must still determine whether the output is stable, externally meaningful, and appropriate for the scientific hypothesis.

2 What we mean by an analytical vector store

An analytical vector store is not defined by the inclusion of a particular clustering algorithm or visual interface. We define it by four properties.

A common corpus. Retrieval and analytical operators refer to the same observations, identifiers, vector representation, and versioned metadata. A result can

Layer	Research function
Corpus	Persistent vectors, stable identifiers, sparse or dense storage, and scientific metadata.
Representation	Derived working views with recorded lineage and measurable distortion.
Operators	Similarity retrieval, cohort operations, clustering, hierarchy, and prospective density or embedding methods.
Validation	Exact or stronger references where affordable, stability analysis, null models, external variables, and negative controls.
Provenance	Versioned parameters, seeds, software state, row coverage, and exportable results.

Table 1: Conceptual layers of an analytical vector store. The definition is method-independent: a proposed implementation is acceptable only if its approximations and result lineage can be evaluated.

therefore be traced back to the stored rows from which it was computed.

Reusable analytical representations. The store may maintain compact auxiliary representations to make wide-vector operations tractable. The research question is whether representations already validated for one task can be reused for other tasks, subject to a new task-specific evaluation. Reuse is a hypothesis to be tested, not an assumption that retrieval accuracy automatically implies analytical validity.

Population operators. The system acts on sets as well as individual queries. Relevant operations include cohort selection, clustering, recursive subdivision, comparisons between subsets, and eventually density and manifold methods. These operators must preserve complete row accounting: every included observation is either assigned, excluded by an explicit rule, or reported as unresolved.

Validation and provenance. An analytical result includes the metric, corpus version, transformation, parameters, random state, stopping rule, and relevant reference calculation. The output is an artefact that can be examined and repeated, rather than an unrecorded state of an interactive session.

3 Why ultra-high dimension requires separate study

Dimensionality is not only a storage concern. It changes which computations are affordable and how geometric measurements should be interpreted. For N observations in D_{raw} dimensions, direct pairwise comparison is $O(N^2 D_{\text{raw}})$ and a Lloyd assignment step is $O(N K D_{\text{raw}})$

for K clusters. These expressions become prohibitive when D_{raw} reaches millions, even if each row is sparse.

Sparsity helps, but nominal dimension still matters. Feature identities may be scientifically meaningful; supports may overlap only weakly; and two representations of the same specimen can have very different sparsity and distance distributions. Concentration effects also make an unqualified distance or inertia difficult to compare across dimensions and support sizes.

Classical dimensionality reduction establishes that suitably constructed projections can preserve finite-set geometry under stated assumptions [2]. Related results show that reduced representations can support k -means under particular objectives and constructions [3, 4]. These results justify investigation, but they do not validate every derived representation or every iterative analytical operator. Our programme therefore treats the following as empirical questions:

1. Which aspects of neighbourhood and population geometry survive a computationally tractable representation?
2. How does fidelity vary with corpus size, sparsity, intrinsic structure, and the scientific metric?
3. When does reuse across retrieval and analysis remain adequate, and when is a task-specific representation required?
4. Which statistics remain comparable as nominal dimension and support change?

4 Research propositions

The work is organised around five propositions. Each is falsifiable on a new corpus and none depends on a claim that a single method is universally superior.

P1: native width need not determine analytical accessibility. For sparse scientific vectors, the cost of routine retrieval and population analysis should depend primarily on corpus size, active support, and chosen working precision, rather than on a requirement to materialise every nominal coordinate in every computational stage.

P2: validated computational structures may support more than one operation. A representation built for similarity search may contain sufficient geometry for selected population analyses, and for prediction tasks such as recommendation or matrix completion. If so, reuse can reduce duplicate computation and keep retrieval, analysis, and prediction attached to one corpus. The proposition must be tested separately for neighbour recall, clustering behaviour, hierarchy, recommendation accuracy, and any future operator; adequacy on one task does not transfer to another.

P3: approximation should be bounded by evaluation. Approximation is often necessary for scale, but it should enter through an identified boundary. Retrieval can be measured against exact neighbours on a defined query set. Clustering can be compared with a direct-space reference where affordable and assessed for stability where it is not. Resource measurements should report cold and warm conditions separately.

P4: discovered structure requires a named reference. A cluster score has no general scientific interpretation without a null model, baseline, or external variable. The appropriate reference may preserve marginal feature frequencies, support size, batch structure, or other domain properties. Different references answer different questions. A numerical separation from one null does not establish a universal effect size or biological class.

P5: multi-scale views are instruments for inspection. Hierarchies and visual maps can expose population, depth, branch structure, and candidate distinguishing variables. They are valuable for audit and hypothesis generation. They do not demonstrate that a partition is chemically, biologically, or clinically valid.

5 Current lines of work

5.1 Similarity retrieval in wide sparse spaces

The first line of work concerns retrieval when direct native-space indexing is expensive. We study families of compact working representations, ensemble retrieval, sparse source storage, and verification against the source vectors. The scientific requirement is a measured recall-latency-memory trade rather than a claim of exact search. Evaluation uses held-out queries and exact brute-force references where corpus size permits.

The present prototype has been exercised on public or research corpora ranging from conventional embeddings to sparse molecular fingerprints with nominal dimension in the tens of millions [11]. These experiments establish feasibility on the tested corpora. They do not establish that the approach dominates specialised sparse indexes or is appropriate for every metric.

5.2 In-store clustering and hierarchy

The second line asks whether the computational structures used for retrieval can also support spherical clustering without exporting the corpus to a separate pipeline. Spherical k -means is a natural reference for unit-normalised vectors and cosine similarity [5]. Mini-batch and coresnet methods address related scaling problems in the number of observations [6, 7].

Our current work studies assignment in compact representations while retaining population summaries in the

source feature space. It includes flat partitions, recursive binary subdivision, bounded-memory execution, explicit stopping rules, and complete row accounting. Direct-space spherical Lloyd is used as an algorithmic reference where affordable; it is not an oracle for a single global optimum. Current molecular hierarchies demonstrate execution at scale but have not yet been validated as chemical or biological taxonomies [12].

5.3 Quality, stability, and null models

Generic cluster scores are often reported without considering dimension, sparsity, or how a centroid was selected. We are studying diagnostics defined relative to explicit generative or resampling references, together with seed stability, subsampling stability, and external-label evaluation. The null should preserve the nuisance structure relevant to the question; an isotropic null, a support-matched null, and a feature-frequency-preserving null are not interchangeable.

This line is deliberately separate from the clustering engine. Fast computation of a hierarchy is a systems result. Evidence that its branches correspond to a replicable scientific distinction is a statistical and domain result.

5.4 Multi-scale inspection

A large hierarchy cannot be evaluated from a list of cluster identifiers. We are investigating renderings that retain tree topology, population, depth, representative observations, and variables associated with each branch. The purpose is to let a domain scientist inspect how a population was divided and select hypotheses for formal testing.

Visualisation introduces its own distortions. Layout distance, area, colour, and branch length must have declared meanings. A visually prominent group may be a layout consequence, and a smooth manifold view may conceal unstable local relationships. We therefore treat the rendering as a view of computed objects, not an independent validation method.

5.5 Manifold embedding

A fourth line asks whether the reused keys support *manifold embedding* — a two-dimensional layout for inspection — as a store operation rather than an exported pipeline. We implement UMAP [9] directly over the retrieval substrate: the neighbour graph is the store’s own approximate k -NN, verified against an exact reference, and the low-dimensional layout is optimised from that graph. Two properties matter for a store. First, the layout is produced deterministically, so the same corpus and parameters yield the same map. Second, a *transform* operation places new observations onto an existing map at retrieval latency, without re-fitting — the neighbourhood primitive is a query, so a point is embedded by

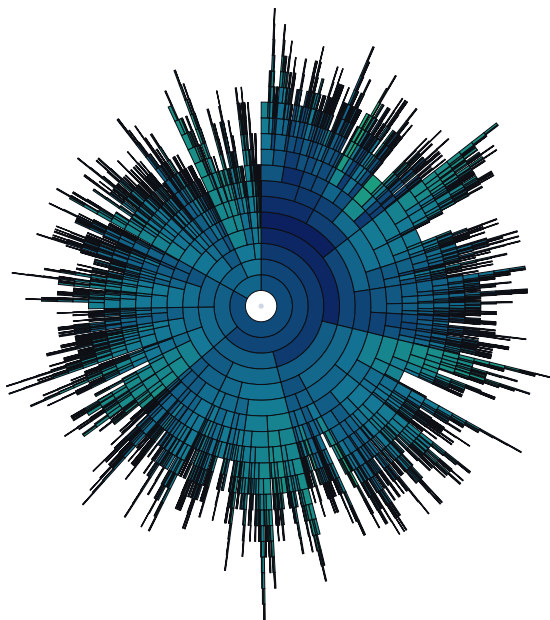


Figure 1: Multi-scale inspection of ultra-wide, 30-million-dimension data by hierarchical clustering. The radial sunburst shows the full chemotype tree computed over 500,000 ChEMBL molecules, each embedded as a wide sparse vector. Every ring is one level of the hierarchy; arc length is proportional to branch population, and colour encodes branch identity. The figure is a rendering of the computed cluster objects, not an independent validation of them.

locating its neighbours among the fitted rows and placing it among their coordinates.

Two quantities are measured separately because they answer different questions, and the distinction is essential to reading such a map correctly. Whether the embedded neighbours of a new point are its *true* high-dimensional neighbours — a rank-based fidelity we can state — is high and roughly independent of width across the corpora tested. How much of a high-dimensional neighbourhood *survives* a projection into two dimensions is a separate, corpus-dependent quantity: it is governed by the intrinsic dimensionality of the data, not by the embedding procedure, and for intrinsically high-dimensional corpora most of the neighbourhood cannot fit in the plane. A two-dimensional separation is therefore a view of computed neighbour relations, not evidence of a discrete population; the quantitative results and their bounds are reported in the companion study [11].

A fitted map need not stay frozen as the corpus grows. We have implemented and evaluated *incremental maintenance*: small batches of arrivals are absorbed into the layout by a micro-update, so the map tracks a growing corpus without a re-fit from scratch. Measured against a from-scratch oracle re-fit on the identical final corpus, the maintained map matches oracle neighbour-rank fidelity

(trustworthiness@15 within 0.017) at about 1.5% of a re-fit’s cost per update. A per-row mobility temperature lets established structure resist new data, so the disturbance of long-fitted rows is bounded (worst-fold aligned residual below 0.04 against a gate of 0.1). Because each update retains its predecessor as an immutable lineage state, drift becomes measurable as *emergence*: the same queries answered against a frozen baseline state and the current state disagree monotonically with injected distributional change, so the maintenance machinery supplies a drift instrument as a by-product rather than merely a fast update. These results are on one wide-sparse corpus with synthetic drift calibration; the detailed protocol and bounds are in the companion study [11].

In use, these properties combine into a monitoring instrument. A growing corpus can be watched on a single living map: arrivals are placed at retrieval latency, each with a measured novelty score derived from its fitted neighbourhood, so routine and unusual arrivals separate on sight; small batches fold into the layout at a small fraction of a re-fit’s cost, so the map tracks the corpus rather than a snapshot of it; and because every update preserves its predecessor, the same queries can be replayed against any frozen baseline for retrospective comparison. Figure 4 shows this operation on the dense corpus of Figure 3: nearly a million fitted articles with a stream of arrivals placed and coloured by measured novelty. As throughout, the display is a view of computed quantities — the novelty score is a measured neighbourhood angle, and whether a red arrival is scientifically anomalous remains a domain judgement.

5.6 Recommendation and matrix completion

A fifth line asks whether the same retrieval-validated representations support *recommendation* on observational matrices, where the task is to predict a held-out interaction or value rather than to retrieve a neighbour. We study two forms of reuse, evaluated against the standard recommender-systems protocol on a public benchmark rather than against a retrieval oracle, so the result is comparable to published collaborative-filtering work.

The first form fits a single linear decoder over the stored keys to reconstruct held-out matrix entries (a matrix-factorisation view in which only the column-factor is learned and the row-factor is the frozen index representation). The second form uses no fitting at all: it answers a recommendation directly from the store’s own nearest-neighbour search, taking the query row’s most similar rows and aggregating their observed interactions. The contrast between the two is itself a finding: a single linear map over a frozen basis is bounded by what that basis encodes, whereas a direct neighbourhood query exploits population co-occurrence the linear map cannot express. On the public benchmark the parameter-free neighbourhood method is the stronger of the two and

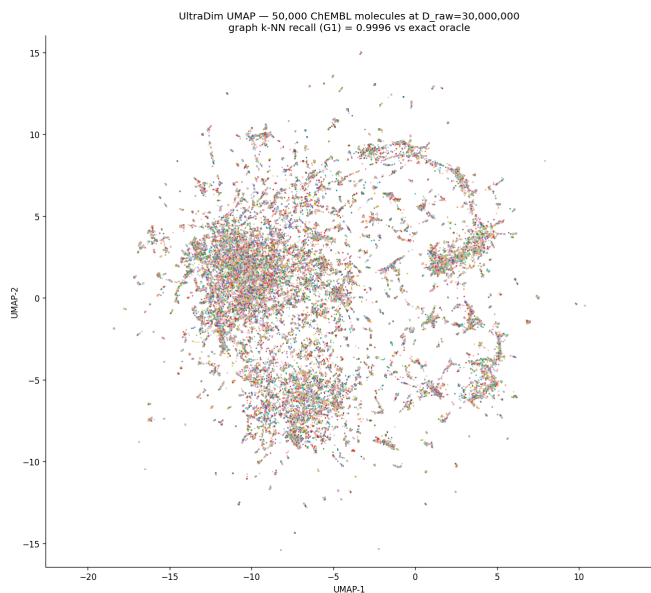


Figure 2: Manifold embedding at ultra-wide scale. A two-dimensional UMAP layout of 50,000 ChEMBL molecules at a nominal width of *thirty million* dimensions ($D_{\text{raw}} = 3 \times 10^7$, sparse Morgan fingerprints), computed entirely from the store’s own approximate neighbour graph (recall 0.9996 against an exact reference) and coloured by the hierarchical-clustering leaf. No neighbour graph is built in the native space; the layout reuses the same keys the index serves search from. The map is a rendering of computed neighbour relations, and how much of each molecule’s high-dimensional neighbourhood a two-dimensional layout can retain is bounded and reported in the companion study, not asserted by the picture.

is competitive with trained baselines, while remaining a query against the live store—so business or scientific constraints expressed as metadata filters are applied during the search itself, not as a post-hoc filter of a fixed model output. As with clustering, this is reuse to be tested per task: the recommendation metric is measured separately from retrieval recall, and the methods are not claimed to dominate trained models. The detailed study is reported separately [13].

5.7 Prospective operators

Density estimation, density-connected clustering, and joint analysis of vector and tabular covariates remain research directions; temporal comparison is now measured for the manifold embedding (§5) but open for the other population operators. DBSCAN-like methods define clusters through local density connectivity [8], and still require the store to supply a density primitive at extreme width. The open problem is not merely to run these algorithms, but to determine whether a persistent store can supply their neighbourhood and population primitives at extreme width while preserving enough information

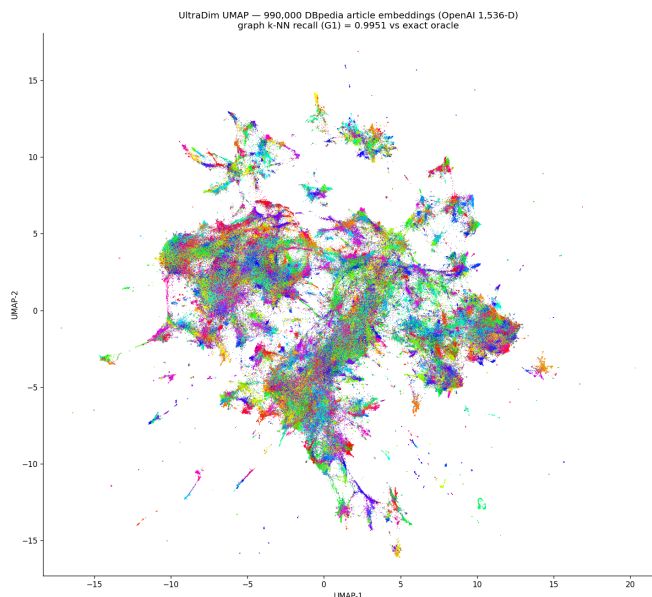


Figure 3: The same operation on a dense corpus. A two-dimensional UMAP layout of 990,000 DBpedia article embeddings ($D_{\text{raw}} = 1,536$), from the store’s own neighbour graph (recall 0.995) and coloured by an in-store k -means partition. At nearly a million points the layout is dense; the fraction of each article’s neighbourhood that survives the projection is limited by the intrinsic dimensionality of the embeddings and is reported, not concealed, in the companion study.

to evaluate the result. Manifold embedding (§5), which posed the same question for neighbourhood relations, has moved from this list to an implemented and evaluated operator; the remaining directions are stated here as open.

6 Scientific domains of interest

The programme is organised around data geometry rather than a single discipline. Three classes of scientific workload are of particular interest.

Molecular representations. Sparse fingerprints and learned molecular representations support similarity, analogue retrieval, library organisation, and exploratory structure–activity analysis. A useful study would compare alternative molecular representations, measure retrieval against chemically defined references, and test whether clusters or branches enrich for held-out scaffold, target, assay, or property labels. The analytical store supplies computation and provenance; chemical interpretation remains a domain evaluation.

Genomic and epigenomic profiles. Whole-genome assays can produce very wide profiles with complex missingness, batch structure, and confounding. Potential questions include cohort retrieval, outlier detection, and

Area	Present status	Principal evaluation	Interpretation boundary
Similarity retrieval	Prototype evaluated on dense and sparse corpora	Recall against exact neighbours; latency; memory; cold/warm behaviour	High recall on one corpus does not establish suitability for another metric or data distribution.
Flat clustering	Implemented and compared with direct-space spherical Lloyd where affordable	Objective value; assignment agreement; seed and sample stability; resource use	Agreement with one initialisation is descriptive because the objective is non-convex.
Hierarchical segmentation	Implemented for wide sparse corpora with complete row accounting	Coverage; stopping behaviour; topology stability; enrichment against held-out variables	A complete or visually separated tree is not a validated scientific taxonomy.
Recommendation / matrix completion	Prototype evaluated on a public benchmark via key-reuse decoder and direct neighbourhood query	Recall and ranking metrics under the standard strong-generalization protocol; comparison with trained and untrained baselines; query latency	Competitiveness on one benchmark, and a parameter-free method, do not establish superiority over trained recommenders or generalisation to other matrices.
Null-relative diagnostics	Experimental	Calibration under simulated and empirical nulls; sensitivity to sparsity, feature frequency, and selection	A result is conditional on the stated null and is not automatically a p -value.
Manifold embedding (UMAP)	Implemented over the reused keys, with out-of-sample transform and incremental micro-batch maintenance of the fitted map	Rank fidelity of embedded neighbours vs. an exact reference; neighbourhood survival vs. intrinsic dimension; deterministic reproducibility; layout vs. an established implementation; incremental map vs. a from-scratch oracle re-fit (quality, cost, old-row stability)	Two-dimensional separation is not evidence of a discrete population, and neighbourhood survival in the plane is bounded by intrinsic dimension. Incremental maintenance is certified on one wide-sparse corpus; its drift detection floor is corpus-specific.
Density-connected methods	Prospective	Neighbour and density fidelity; stability; comparison with established implementations	A density primitive at extreme width remains to be supplied and evaluated.

Table 2: Current scope of the research programme. An implemented component is a working research prototype, not evidence of production maturity or domain validation.

multi-scale patient or sample stratification. Any study must pre-specify preprocessing, account for batch and clinical covariates, and separate exploratory subgroup discovery from prognostic or diagnostic claims. The present system is a research instrument and is not a medical device.

Large observational matrices. Customer-product, specimen-feature, and event-attribute matrices provide useful stress tests for sparse computation and population analysis. They are also methodologically valuable because known temporal splits or held-out outcomes can be used to test stability and generalisation, and because the held-out-entry prediction task is a standard, externally comparable benchmark for recommendation and matrix completion (§5). A computationally coherent partition, or a competitive recommendation score, is not necessarily an actionable or causal segmentation.

Other domains are relevant where observations admit a scientifically defensible vector representation and metric. A very wide vector is not sufficient by itself: the research value depends on whether neighbourhood, pop-

ulation, and external validation questions can be stated before inspecting the output.

7 Evaluation principles

An early-access study should begin with an evaluation design, not with a preferred picture or partition. We use the following hierarchy of evidence.

- 1. Representation checks.** Document feature construction, scaling, normalisation, missing values, sparsity, and the scientific meaning of the distance metric. Compare plausible representations where the choice is not settled.

- 2. Computational fidelity.** Measure approximate retrieval against exact search on a defined sample. Where direct clustering is affordable, compare objectives and assignments under matched initial conditions. Report coverage, failures, memory, and wall time.

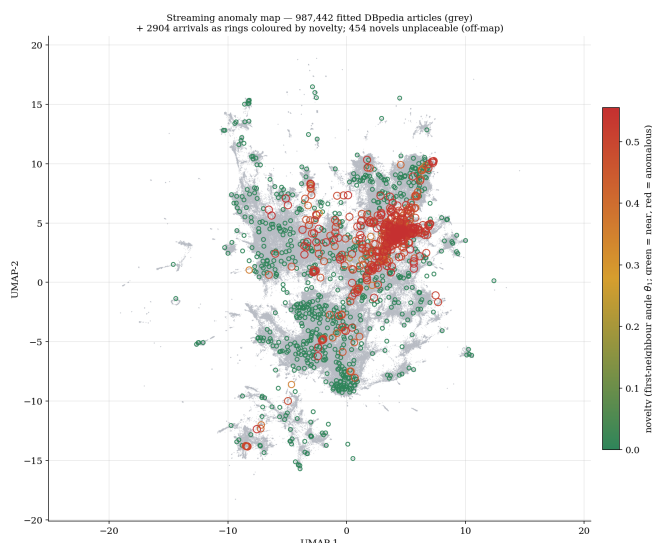


Figure 4: Streaming arrivals on a living map. The fitted layout of 987,442 DBpedia articles (grey) with 2,904 arrivals placed by the transform operation and ringed by measured novelty: green arrivals land in well-supported neighbourhoods, red arrivals do not, and 454 arrivals that could not be placed are reported off-map rather than forced onto the layout. Placement happens at retrieval latency, and the maintained map absorbs arrival batches at a small fraction of a re-fit’s cost, so the view stays current as the corpus grows. The novelty score is computed from each arrival’s fitted neighbourhood; anomaly in the scientific sense remains a domain judgement.

3. Robustness. Repeat across random state, subsamples, corpus order, and reasonable parameter ranges. For a hierarchy, assess branch and membership stability rather than only the number of leaves.

4. Statistical reference. Choose nulls and negative controls that preserve known nuisance structure. Separate exploratory reuse of the same observations from confirmatory evaluation on held-out data. Avoid interpreting an adaptively selected cluster with a test that assumes it was specified in advance.

5. Domain validation. Relate results to independent labels, assays, outcomes, interventions, or expert annotations. Report both positive and negative findings. Domain validation is the stage at which a computational structure may acquire scientific meaning.

6. Reproducibility. Record corpus and software versions, parameters, seeds, row inclusion, intermediate artefacts, and evaluation code. These requirements follow the wider principle that computational results should be findable, accessible, interoperable, and reusable where governance permits [10].

8 Academic early access

The academic early-access programme exists to evaluate the research propositions on real scientific workloads and to expose cases in which they fail. It is not a production service, a clinical decision system, or a substitute for peer review. The intended participant is a research group able to contribute domain knowledge, a defined analytical question, and an independent basis for evaluation.

8.1 A suitable project

A project is a strong candidate when:

- the corpus is high-dimensional, large, sparse, or expensive to analyse repeatedly with current tools;
- the vector representation and similarity metric have a defensible domain interpretation;
- there is a concrete retrieval, cohort, clustering, hierarchy, or exploratory visualisation question;
- baseline methods or exact calculations are available on at least a representative subset;
- held-out labels, outcomes, assays, or expert annotations can evaluate scientific meaning; and
- the group can work with research software whose interfaces and performance characteristics may change.

A project is not yet suitable when the main objective is regulated clinical use, when no meaningful distance or validation target can be stated, or when the required conclusion depends only on the visual appearance of an embedding. Moderate-dimensional workloads already served well by established tools may still provide useful controls, but are not the primary research target.

8.2 Study sequence

An early-access study is expected to follow five stages:

- 1. Non-confidential scoping.** Define the scientific question, corpus geometry, candidate metrics, comparison methods, and governance constraints. Aggregate dimensions and sparsity statistics are usually sufficient for this stage.
- 2. Evaluation design.** Agree the reference calculations, external variables, negative controls, resource measurements, and criteria for a useful or negative result.
- 3. Technical trial.** Ingest a public, synthetic, de-identified, or institutionally controlled corpus as appropriate, and verify retrieval, row accounting, and reproducibility before interpreting analytical outputs.
- 4. Domain analysis.** Examine stability and external validity with the participating scientists. Exploratory findings are labelled as such.

5. **Research report.** Record methods, results, failures, limitations, and next experiments. Publication and data release are separate decisions governed by the participating institutions.

Data location, access, retention, and disclosure must be agreed for each project. The initial decision to participate does not require transfer of a scientific dataset. Work involving patient or otherwise sensitive data proceeds only under the relevant institutional approvals and data-governance arrangements.

8.3 What the programme seeks to learn

The research team is seeking evidence on questions that cannot be answered by systems benchmarks alone:

- Which scientific representations retain useful neighbourhood and population structure under compact computation?
- Which failure modes arise from sparsity, batch effects, feature frequency, or heterogeneous support?
- Which null models and stability measures are credible in each domain?
- Which analytical outputs change a scientist’s next experiment, and which merely restate known structure?
- What provenance and interaction are required for a domain expert to audit a large hierarchy or population map?

Negative results are part of the programme. A finding that a representation, metric, operator, or visualisation is unstable narrows the valid scope of the system and may be more informative than another favourable benchmark.

9 Open research questions

Representation dependence. Neighbourhood and cluster structure can change when the feature map, metric, or normalisation changes. We need methods for comparing representations without assuming that any one of them is a ground truth.

Adaptive analysis. Interactive subdivision, parameter selection, and visual inspection reuse the same observations repeatedly. Conventional inferential quantities may be miscalibrated after this adaptive search. Confirmatory claims require held-out data, selective-inference methods, or an explicitly exploratory status.

Heterogeneous sparsity. Support size may vary systematically with experimental protocol or population. A null matched only on nominal dimension can then produce misleading cohesion or density. Reference models must account for the nuisance structure that the scientific design creates.

Temporal and corpus drift. Persistent stores change. New observations may alter neighbourhoods, centroids, and branch boundaries. For the manifold embedding this is now an implemented and evaluated operator: incremental maintenance updates a fitted map from arriving batches, and its retained lineage measures analytical drift as emergence rather than supplying merely a fast update (§5). The corresponding incremental methods for centroids and branch boundaries, and their drift measures, remain open.

Uncertainty in visual summaries. Current visualisations generally show one fitted structure. A scientifically useful display should also communicate unstable membership, alternative partitions, missing observations, and sensitivity to the chosen representation.

External validity. Computational agreement and stability are necessary but insufficient. Generalisability across sites, instruments, populations, or chemical series must be tested with independent data and domain-specific study designs.

10 Limitations of the present programme

The current evidence comes from a limited collection of public and derived corpora and predominantly from one hardware environment. Some evaluations use point measurements rather than intervals over repeated datasets or systems. Specialised sparse retrieval and clustering methods are relevant baselines and have not yet been matched across every extreme-width workload.

The implemented clustering operators are approximate and iterative. Their behaviour is measured empirically; the complete operators are not covered by a general convergence or approximation theorem. Hierarchical output is sensitive to representation, initialisation, stopping rules, and minimum population.

Current quality diagnostics are conditional on specified null models. They are not universal measures of cluster validity, calibrated significance, or biological effect. Current molecular results have not established activity, target, scaffold, toxicological, or clinical validity. Proposed genomic and epigenomic applications require separate studies and appropriate governance.

The early-access programme is consequently a research collaboration. It is designed to establish where the analytical-vector-store model is useful, where it needs modification, and where established alternatives remain preferable.

11 Conclusion

An analytical vector store is a proposal for organising scientific computation: store the vector corpus once; attach

retrieval and population analysis to that corpus; reuse validated computational representations where the evidence supports reuse; and preserve the lineage required to evaluate every result. The approach is motivated by ultra-high-dimensional sparse data, but its scientific value depends less on nominal scale than on explicit metrics, references, stability tests, and external validation.

Our present work covers wide-vector retrieval, spherical clustering, recursive segmentation, recommendation and matrix completion, null-relative diagnostics, and multi-scale inspection. Density, manifold, temporal, and covariate-aware methods remain open research areas. The academic early-access programme provides a route for laboratories to test these ideas against real scientific questions under a pre-specified evaluation design. Its purpose is to produce evidence about the scope and limitations of analytical vector stores, including evidence that particular methods or workloads do not fit the model.

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